

THE CONCLUSION



01. LEAN TOOL: CONTINUOUS FLOW

Continuous Flow

Takt Time

Pull System

Kanban

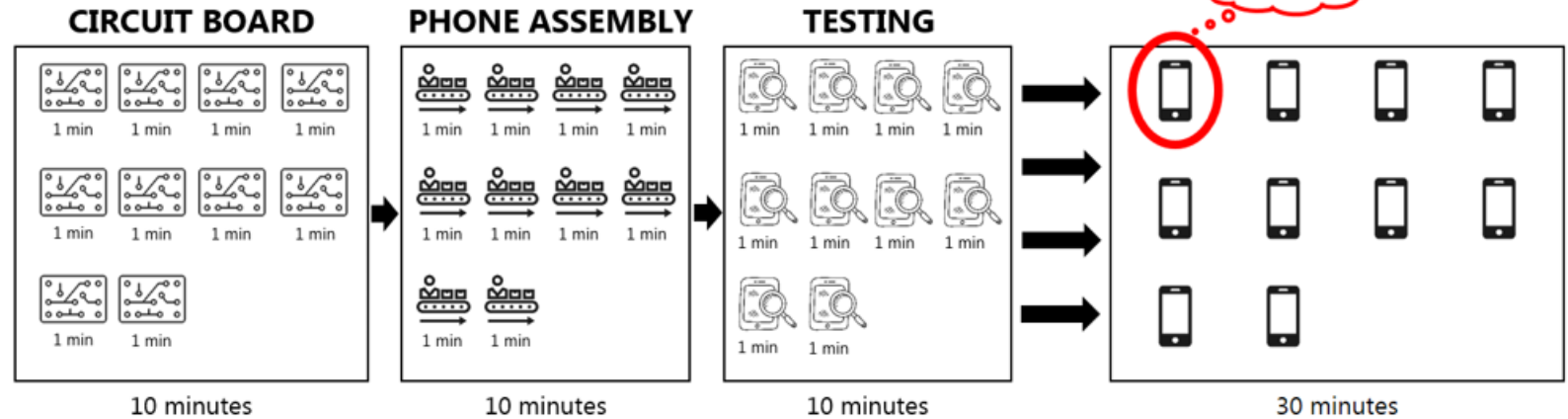
Heijunka

Standard Work

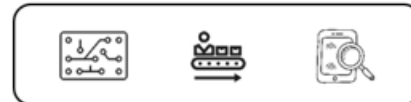
Visual Control

5 Why Analysis

EXAMPLE



ONE-PIECE FLOW CELL



= just **3 minutes** instead of 21 to make the first smartphone

= just 12 minutes to make 10 smartphones, while the batch flow process takes 30 minutes

Operators were not allowed to build up inventory

Flow has **eliminated overproduction and inventory**

02. LEAN TOOL: TAKT TIME

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

WHAT IS TAKT TIME?

AIGPE

Time needed to produce one unit of a product to meet customer requirements

$$T = \frac{T_a}{T_d}$$

where, T stands for Takt Time

T_a stands for Net Available Time

T_d is the total customer demand (units produced per day)

German word for
“Drum beat”

Establishes the pace of
the process needed to
meet customer demand

Shows where the flow
is uneven

Takt time can be calculated after creating high level, functional process maps, and value stream maps

03. LEAN TOOL: PULL SYSTEM

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

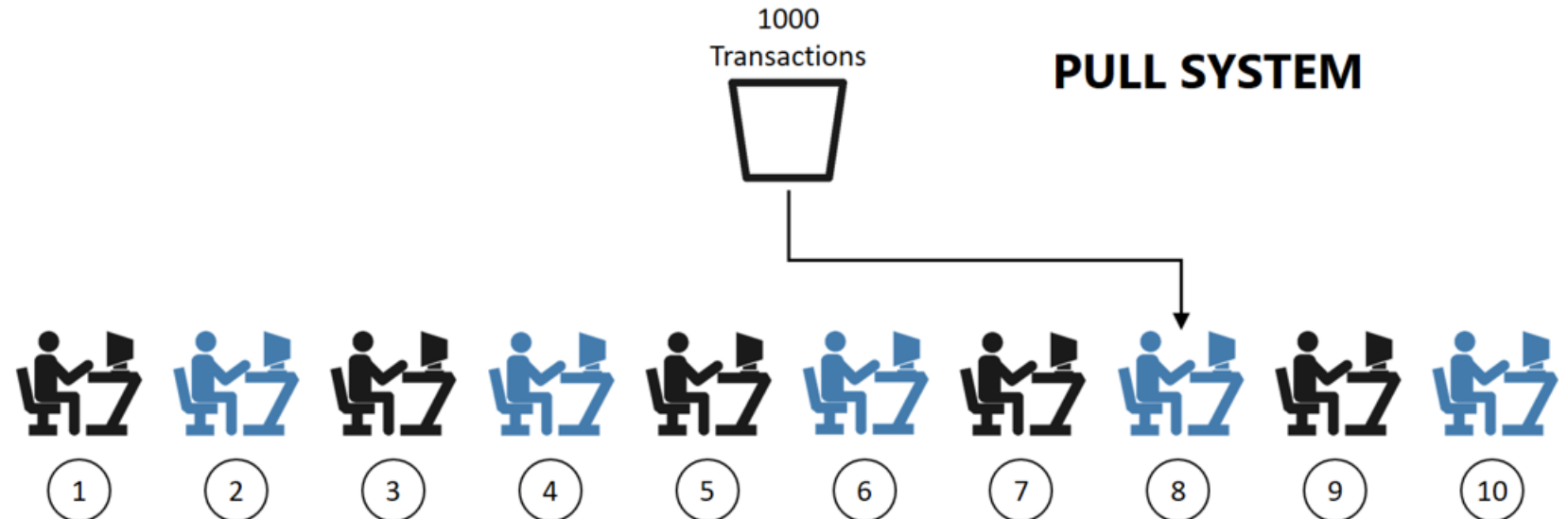
Standard Work

Visual Control

5 Why Analysis

EXAMPLES OF PULL SYSTEM

AIGPE



THIS KIND OF A PULL SYSTEM WILL ALLOW A HEALTHY COMPETITION

IT WILL ALSO ELIMINATE THE PROBLEMS ASSOCIATED WITH THE PUSH SYSTEM

04. LEAN TOOL: KANBAN

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

AIGPE

WHAT IS KANBAN?

FACTORIES CAN BE LARGE AND SPREAD OUT

SUPPLIERS OF PARTS ARE GENERALLY A DISTANCE AWAY

TAIICHI OHNO LOOKED FOR A WAY TO SIGNAL THAT THE ASSEMBLY LINE HAD USED THE PARTS AND NEEDED MORE



CARDS



EMPTY BINS



EMPTY CARTS

THIS IS CALLED "KANBAN"

05. LEAN TOOL: HEIJUNKA

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

EXAMPLE – TRADITIONAL PRODUCTION (UNLEVELED)

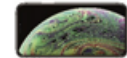
AIGPE

Monday
Production

iPhone XS



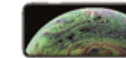
iPhone XS



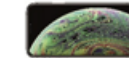
iPhone XS



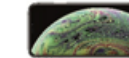
iPhone XS



iPhone XS



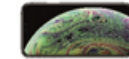
iPhone XS



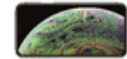
iPhone XS



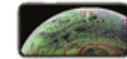
iPhone XS

Tuesday
Production

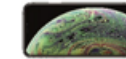
iPhone XS



iPhone XS



iPhone XS



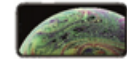
iPhone XS



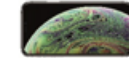
iPhone XS



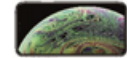
iPhone XS



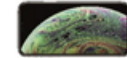
iPhone XS

Wednesday
Production

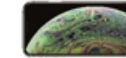
iPhone XS



iPhone XS



iPhone XS



iPhone XS

Changeover



iPhone XR



iPhone XR

Thursday
Production

iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR



iPhone XR

Friday
Production

iPhone XR

Changeover



iPhone XS Max



iPhone XS Max



iPhone XS Max



iPhone XS Max

06. LEAN TOOL: STANDARD WORK

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

WHAT IS STANDARD WORK?

AIGPE

**STANDARD WORK IS A TOOL THAT
DEFINES THE INTERACTION BETWEEN...**



MAN

&



MACHINE

HAS 3 COMPONENTS

**STANDARD
TIME**

**STANDARD
INVENTORY**

**STANDARD
SEQUENCE**

**STANDARD WORK HELPS IN TRAINING NEW OPERATORS AND
REDUCING THE VARIATION IN THE PROCESS**

07. LEAN TOOL: VISUAL CONTROL

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

WHEN IS VISUAL CONTROL USED?

AIGPE

CAN BE USED AS CREATIVELY AS POSSIBLE



**KEEP YOUR SOILED
PLATES HERE**



08. LEAN TOOL: 5 WHY ANALYSIS

Continuous Flow

Takt Time

Pull System

Kanban

Heijunka

Standard Work

Visual Control

5 Why Analysis

AIGPE

5 WHY EXAMPLE 02

One of the monuments in Washington D.C. is deteriorating!

- | | | |
|---|--|---|
| 1 | Why is the monument deteriorating? | Because harsh chemicals are frequently used to clean the monument |
| 2 | Why are harsh chemicals needed? | To clean off the large number of bird droppings on the monument |
| 3 | Why are there a large number of bird droppings on the monument? | Because the large population of spiders in and around the monument are a food source to the local birds |
| 4 | Why is there a large population of spiders in and around the monument? | Because vast swarms of insects, on which the spiders feed, are drawn to the monument at dusk |
| 5 | Why are swarms of insects drawn to the monument at dusk? | Because the lighting of the monument in the evening attracts the local insects |

Solution: Change how the monument is illuminated in the evening; prevent attraction of insects

LEAN MANAGEMENT – PART 02

What is Continuous Process Flow?

Section Introduction

What is Continuous Process Flow?

Traditional Production vs Continuous Flow

Benefits of Creating Continuous Flow

Activity: True or False

Activity Solution: True or False

Section Conclusion

What is Takt Time?

Section Introduction

What is Takt Time?

Formulate to Calculate Takt Time

Activity: Calculate the Takt Time

Activity Solution: Calculate the Takt Time

Section Conclusion

What is Pull System?

Section Introduction

What is Pull System?

Examples of Pull System

Activity: Use Pull System for Given Situations

Activity Solution: Pull System for Given Situations

Section Conclusion

What is Kanban?

Section Introduction

What is Kanban?

Examples of Kanban in Everyday Life

Steps to Implement Kanban

Activity: Identify the Kanban System

Activity Solution: Identify the Kanban System

Section Conclusion

What is Heijunka?

Section Introduction

What is Heijunka?

Role of Muda, Muri and Mura

Steps to Implement Heijunka

Heijunka in Service Operations

Activity: Heijunka – True or False

Activity Solution: Heijunka – True or False

Section Conclusion

What is Standard Work?

Section Introduction

What is Standard Work?

Is Standard Work Basis of all Improvements?

Key Points to Remember – Standard Work

Section Conclusion

What is Visual Control?

Section Introduction

What is Visual Control?

Steps to Implement Visual Control

Section Conclusion

What is 5 Why Analysis?

Section Introduction

What is 5 Why Analysis?

Real-Life Examples of 5 Why Analysis

Steps to Implement 5 Why Analysis

Activity: Use 5 Why Analysis

Activity Solution: Use 5 Why Analysis

Section Conclusion

Section

Lecture

Activity

CELEBRATE

AIGPE





Go out for lunch or dinner



**Watch another Marvel or DC
Superhero Movie**



**Share your selfie with a quote "*I
am a Lean Specialist*"**

THANK YOU FOR ATTENDING...
SEE YOU IN THE NEXT COURSE...